

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2023 - present	B.Tech.	Indian Institute of Technology Kanpur	8.8/10.0
2023	CBSE(XII)	Lord Budhha Public School, Kota	94.8%
2021	CBSE(X)	St.Thomas Sr. Sec. School, Chitrakoot	97.6%

SCHOLASTIC ACHIEVEMENTS

- Received the **Academic Excellence Award** for exceptional performance in academics during 2024–2025 at IIT Kanpur. (2025)
- Secured **2nd Runner-Up** at the **Regional Round** of the **Bharat Nav-Nirmaan Challenge** and **qualified for the Nationals**. (2026)
- Qualified for **NTSE Stage 1** conducted by **NCERT** among two lakh candidates appearing all over India. (2021)
- Achieved **A*** or **A** grade in **17 courses** spanning Data Science, Computer Vision, Civil Engineering, and Mathematics.

RESEARCH EXPERIENCE

Exploration of topic modeling frameworks for crash narrative analysis SURGE Prof. Aditya Medury, IITK  (May'25–July'25)	
Objective	- Understand data generation process used within Latent Dirichlet Allocation (LDA) and Structured Topic Model (STM). - Assess the role of metadata in STM in improving topic modeling outcomes compared to conventional LDA frameworks. - Applying these techniques to crash narratives using Michigan crash data, U.S.A., for topic extraction and interpretation.
Approach	- Reviewed probabilistic foundations of LDA/STM, emphasizing Dirichlet, multinomial, and logistic-normal distributions. - Pre-processed crash narratives using lemmatization and stop word removal to reduce noise and standardize vocabulary. - Implemented LDA and STM for topic modeling, with STM incorporating metadata using the crash narrative data.
Observation	- Structured metadata (e.g., highway class, lighting conditions) were used to model topic proportions for each document in LDA, and their relationship with topics was later interpreted using regression techniques like Beta regression. - Identified key latent topics in crash narratives, including loss of control, animal-involved collisions and among others.

WORK EXPERIENCE

Fraud Alert Consolidation BIU Intern Axis Bank (May'26–July'26)	
Objective	- Consolidate 3 separate fraud datamarts (CC, DC, UPI) and multiple intermediate tables into a single unified Central Fraud Fact Table on a Big Data stack (Hadoop/Hive/HDFS), eliminating redundant I/O across all downstream CPFIR regulatory scripts. - Enable cross-channel fraud analytics and provide a permanent RBI-compliant audit trail while preserving all existing channel-specific fraud detection logic.
Approach	- Analyzed 25+ production PySpark scripts to trace source-to-target data lineage across multiple FICO Falcon source tables and extract embedded ETL and enrichment logic. - Designed an 80-column unified star schema with a composite primary key and channel discriminator using Spark SQL / HiveQL, resolving duplicate rates in upstream keys. - Built the end-to-end ETL pipeline in PySpark, integrating channel-specific fraud detection logic and consolidating customer lookups into a single enrichment step using broadcast joins. - Optimized ingestion of a 20.8M rows/day source by converting TextFile → Snappy Parquet for significant read speedup; implemented append-only partitioned writes with batch-level audit columns for traceability. - Evaluated two design variants — a lean consumer-driven model (~80 columns) and a full datamart replica (~330 columns) — and recommended the lean model based on HDFS quota constraints and downstream query patterns.
Results	- Validated 129.5M+ rows across all three channels using SQL-based checks covering volume, fraud detection flags, enrichment coverage, and data quality. - Projected ~80% reduction in daily I/O and HDFS storage, ~60% in wall-clock runtime and daily Spark joins; authored architecture documentation and presented impact analysis to senior management. - Received a Pre-Placement Offer (PPO) based on internship performance.

CERTIFICATIONS

IIT Kanpur Certification Program on Python  Department of Electrical Engineering, IITK  (Nov'24- Dec'24)	
Objective	Learn hands-on implementation of modern algorithms in AI, ML, Neural Networks, and Deep Learning.
Approach	- Acquired proficiency in core libraries: TensorFlow, NumPy, Keras, Pandas, Matplotlib, and Scikit-Learn. - Executed multiple projects using real-world datasets in domains such as image classification (MNIST, CIFAR-10, Fashion-MNIST), medical diagnostics (Breast Cancer Wisconsin), housing price prediction (Boston Housing). - Studied mathematical concepts behind key algorithms: SVM, Naive Bayes, Decision Trees, and Clustering etc.
Impact	Strengthened understanding of algorithm design and statistical learning workflows for applied data science projects.

KEY PROJECTS

SAMCo: Automatic Object Co-segmentation EE655 Course Project Prof. Koteswar Jerripothula  (Jan'26–Apr'26)	
Objective	Developed an automated zero-shot object co-segmentation framework leveraging Vision Foundation Models (Meta's SAM & DINOv2 ViT) and Semantic Consensus Prompting to segment shared concepts without labels.
Approach	- Extracted 384-dim dense patch tokens from self-supervised DINOv2, computing pairwise cross-image cosine similarity to generate semantic consensus saliency maps. - Formulated an automated prompting strategy converting saliency maps into discrete foreground/background point prompts via spatial K-Means clustering. - Engineered an Iterative Consistency Feedback Loop that recirculates segmented masks to suppress background noise and refine fine-grained object boundaries over successive iterations.
Results	Achieved mIoU of 0.781 (iCoseg) and 0.729 (MSRC v2), outperforming existing baselines and gaining 3.8% mIoU via iterative feedback; evaluated with IoU, Dice, Precision, Recall & MAE.

Objective	Built a robust image classifier on 37k images across 10 classes using WideResNet-28-10 (36.5M params) to handle real-world corruptions (Gaussian noise, motion blur, JPEG artifacts) without pretrained weights or external data.
Approach	- Resolved a 50:1 class imbalance across minority and majority classes using WeightedRandomSampler with inverse-frequency weights to prevent minority class collapse on Macro F1-score. - Enforced corruption-invariant feature learning via AugMix with dual-view Jensen-Shannon Divergence (JSD) consistency loss, Mixup, CutMix, and Stochastic Depth (DropPath). - Designed an iterative pseudo-labeling pipeline with confidence-based filtering, progressively incorporating unlabeled test images into fine-tuning across successive refinement rounds with SWA.
Results	Achieved a public leaderboard Macro F1 of 0.904 via 50-pass Test-Time Augmentation (TTA) ensemble using PyTorch AMP for mixed-precision accelerated training.

Objective	Prepared a sentiment analysis system by integrating text & image data to classify internet memes by sentiment.
Approach	- Implemented DistilBERT for text analysis and Vision Transformer (ViT) for image-based sentiment prediction. - Used PyTorch and Hugging Face Transformers for development and text preprocessing, tuning to enhance accuracy. - Worked with mixed data formats: CSV files, OCR-extracted text, image data and integrating to analyse sentiment.
Impact	Achieved 87% F1-score, outperforming text-only 81% F1-score and image-only 74% F1-score enhancing accuracy.

Objective	Executed a field-to-finish topographic survey of the Lecture Hall Complex (L18–L20) and created an accurate, large-scale digital GIS map with elevation contours for site analysis.
Approach	- Established a 4-station closed traverse using a Trimble S5 Total Station; adjusted angular and linear closing errors using the Bowditch (Compass) Rule. - Conducted a differential levelling circuit via Rise and Fall Method from a standard benchmark to establish station Reduced Levels (RL) with high closing accuracy. - Processed 3D spatial coordinates in QGIS (WGS 84 / UTM Zone 44N, 1:800 scale), digitizing features and generating IDW-interpolated topographic contours.
Result	Delivered a cartographic map of the L20 complex, integrating survey data with GIS layers for terrain gradient analysis and infrastructure visualization.

Objective	Built an autonomous AI research agent that retrieves information from multiple sources, fact-checks its own responses, and produces fully cited, hallucination-free answers using a multi-step agentic pipeline.
Approach	- Designed a multi-step agent pipeline in LangGraph where the agent plans research, decomposes queries, and uses an LLM-as-a-Judge reflection step to evaluate evidence quality before synthesizing a response. - Built a Dual-Channel Retrieval system combining live DuckDuckGo web search with a local ChromaDB vector database to fetch the most relevant context for each query. - Implemented a Grounding & Citation Gate that breaks each response into individual factual claims, cross-checks them against retrieved sources, and rewrites any unsupported statements before returning the final answer.
Result	Deployed a production-ready FastAPI backend supporting Gemini, Groq, and OpenRouter as interchangeable LLM providers, containerized with Docker and validated with 125 unit tests.

Objective	Built RASTA-SAFE — a rule-first AI traffic enforcement platform to reduce road accidents by enabling automated, explainable, and hardware-agnostic violation detection across existing city CCTV infrastructure.
Approach	- Fine-tuned YOLOv8-nano on Indian traffic datasets for real-time detection of signal violations, speeding, and helmet non-compliance across 25+ urban surveillance cameras via existing RTSP CCTV feeds. - Built deterministic rule engines (RLVD + speed thresholding) in OpenCV/Python ensuring AI perception is validated by law-aligned logic before any enforcement action is triggered. - Designed a Corridor Risk Index (CRI) aggregating speed violations, signal breaches, and near-miss events for spatial hazard scoring and data-driven patrol deployment.
Result	Integrated the pipeline with a SQL database, live dashboard, and e-Challan/VAHAN APIs for automated challan dispatch; system design targets 80% lower CapEx over hardware-heavy alternatives.

- Developed a modular **Command-Line Banking System** in **C++** applying core **OOP concepts** and layered architecture.
- Implemented polymorphic account models (**Savings** with interest rate calculation, **Current** with custom overdraft limits).
- Supported essential banking workflows including **deposits, withdrawals, fund transfers**, and detailed **transaction history logs**.
- Built a dual-mode persistence layer supporting lightweight file I/O and a **MySQL backend** with auto-migrating schemas and **ACID transaction integrity**.
- Developed an **Admin Interface** for customer approvals, global interest configuration, and aggregate balance/liability reporting.
- Implemented secure customer authentication with **4-digit PIN hashing** and robust **input validation** to prevent invalid operations.

- Built a high-performance **In-Memory Key-Value Store** in **C++** delivering constant-time $\mathcal{O}(1)$ **GET/SET/DEL** operations using a **Hash Map** and custom **Doubly Linked List**.
- Implemented pluggable **LRU and LFU cache eviction** policies to manage fixed-memory constraints alongside real-time telemetry tracking **cache hit/miss ratios**.
- Designed **Time-To-Live (TTL)** key expiration with asynchronous **background cleanup**, ensuring **thread-safe concurrency** for simultaneous multi-threaded operations.
- Integrated **binary snapshot persistence** for atomic state recovery across restarts and benchmarked throughput reaching **600k+ QPS**.

Optimizing Air Travel Through Flight Delay Prediction Self Project 		
- Analyzed 180k+ US DOT flight records across major airports and airline carriers to uncover temporal, weather, and operational bottleneck trends. - Formulated a novel Operational Adjustability Index (OAI) weighting internal controllable delays $2\times$ over external stochastic factors. - Built Random Forest Classifiers with SMOTE to handle class imbalance, achieving an F1-score of 0.83 on severe delay events. - Trained Random Forest Regressors predicting delay duration per flight, achieving an MAE of 21.1 min and RMSE of 38.5 min.		
RELEVANT COURSEWORK (*Ongoing)		
Data Science & Machine Intelligence(A)	Computer Vision & Deep Learning	Applied Probability & Statistics(A)
Computational Methods in Engineering	Optimization Methods in Engineering*	Linear Algebra
Multivariable Calculus	Ordinary Differential Equations	Partial Differential Equations
Fluid Mechanics(A)	Structural Analysis(A)	Soil Mechanics(A)
Transportation Engineering(A)	Engineering Hydraulics(A)	Environmental Quality & Processes(A)
Civil Engineering Materials(A*)	Engineering Hydrology	Foundation Design
Principles of Geoinformatics	Reinforced Earth Structures	Construction Management
Design of Steel Structures	Mechanics of Solids	Introduction to Electronics

TECHNICAL SKILLS		
Languages & Databases	ML/AI & Big Data Frameworks	Developer Tools & Software
C, C++, Python, SQL, PySpark, HiveQL, R, MATLAB, L ^A T _E X, MySQL	PyTorch, TensorFlow, Keras, LangGraph, LangChain, Hugging Face, YOLOv8, OpenCV, Scikit-learn, Apache Spark (HDFS), NumPy, Pandas	Git/GitHub, Docker, FastAPI, Linux/Bash, QGIS, Autodesk Fusion 360, AutoCAD, Photoshop, Lightroom

POSITIONS OF RESPONSIBILITY	
Senior Executive Media and Publicity Udghosh'24 (Sep'24-Oct'24)	
Leadership	- Successfully collaborated with media representatives from leading platforms to amplify event visibility and outreach. - Effectively led team of 10+ volunteers, ensuring smooth execution of merchandise management and audience engagement.
Operations	- Managed promotional content creation for newsletters and digital campaigns, ensuring consistent event branding. - Seamlessly coordinated influencer appearances and scheduled social media promotions to maximize engagement.
Impact	- Boosted online event reach and audience participation through coordinated influencer and digital marketing efforts. - Boosted merchandise sales and attendee participation, contributing to overall event revenue and reach.
Secretary Photography Club IITK (May'24-April'25)	
Leadership	- Organized multiple photography contests for campus community 9K+, engaging in themed and open-category challenges. - Coordinated and hosted 'PIXELS', a week-long photography workshop for 80+ attendees, featuring professional talks.
Operations	- Conducted Bulla sessions and competitions for Freshers'24, managing event design, judging, and prize distribution. - Worked on the participating team for IFP'24, a nationwide 50-hour photography challenge, handling photo curation.
Impact	- Established a dynamic platform for photography enthusiasts on campus to enhance their Photography skills. - Attracted diverse audience including professors and students from various disciplines in the campus for the workshop.

EXTRA-CURRICULAR ACTIVITIES	
Volunteer	Volunteered as a Junior Executive in the Show Management Vertical at Antaragni'23 coordinating event logistics.
Sports	Participated in the Powerlifting competition organized as part of Freshers Inferno'23, Fresher Sports championship.
Cultural	Took part in the Photography competition during Galaxy'24, the annual inter-hall cultural championship.
Managerial	Led the Hall 5 team Pool Nawabs of 15+ members in the various Photography competition at Galaxy'25.